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Synopsis

Smart Bridge Health Monitoring and Earthquake Detection System

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering
In
Computer Science and Engineering (Data Science)

by

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I/we declare that this written submission represents my/our ideas in my/our own words and where others ideas or words have been included, I/we have adequately cited and referenced the original sources. I/we also declare that I/we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my/our submission. I/we understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

Bridges are essential components of transportation infrastructure, and their safe operation is important for public mobility and economic activity. During service, bridges are continuously exposed to moving-vehicle loads, wind, environmental changes, material ageing, and, occasionally, strong ground motion. Periodic visual inspection remains necessary, but it may not capture short-duration vibration events or changes in structural response that occur between inspections. Structural Health Monitoring (SHM) uses sensor measurements to observe the behaviour of a structure over time and to identify unusual responses that may require further examination. This project proposes a low-cost Smart Bridge Health Monitoring and Earthquake Detection System designed as an educational prototype for monitoring vibration on a model bridge.

The proposed hardware integrates an ESP32-C3 Super Mini microcontroller, an MPU6050 accelerometer and gyroscope module, an ADS1115 16-bit analog-to-digital converter (ADC), and two piezoelectric (PZT) sensors. The MPU6050 measures acceleration and angular motion along three axes, providing information about the motion experienced by the model bridge. The PZT sensors produce electrical signals in response to mechanical stress and vibration at their mounting locations. Since these signals are analog, the ADS1115 converts them into digital samples through its analog input channels. The ESP32-C3 communicates with the MPU6050 and ADS1115 over the I²C bus, processes the measurements, and coordinates the monitoring process.

During operation, the controller acquires sensor readings at selected intervals, applies basic filtering to reduce noise, and calculates simple vibration indicators such as peak amplitude and root mean square (RMS). The measured values are compared with a baseline and thresholds established through controlled testing. Readings within the calibrated range are treated as normal monitoring conditions, while readings that exceed the selected thresholds can trigger an elevated-vibration indication and an event record. Optional Wi-Fi reporting can make readings and status information available for remote viewing, while local monitoring can continue independently of network availability. Using both inertial and piezoelectric sensing provides complementary observations of the model bridge's response and supports comparison between sensor channels.

The project aims to demonstrate sensor interfacing, embedded data acquisition, basic signal processing, and threshold-based event indication using readily available components. The prototype will be evaluated through controlled bench or model-bridge tests, with sensor mounting, signal conditioning, repeatability, and threshold selection considered during testing. Its results are intended to support learning and preliminary anomaly screening, not to provide a definitive diagnosis of structural damage. Earthquake-related detection refers only to identifying strong motion measured by the installed sensors after shaking reaches them; the system does not predict earthquakes or provide certified early warning. It is not a replacement for professional bridge inspection, calibrated seismic instruments, or engineering assessment. The project demonstrates how an affordable, modular embedded system can introduce the principles of bridge vibration monitoring and encourage further work on reliable, data-driven infrastructure monitoring.

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Abbreviations and symbols

- IoT: Internet of Things
- MCU: Microcontroller Unit
- ESP32-C3 Super Mini: Wi-Fi/Bluetooth Low Energy microcontroller development board
- MPU6050: Six-axis MEMS accelerometer and gyroscope module
- ADS1115: 16-bit I²C analog-to-digital converter (ADC)
- PZT: Lead zirconate titanate piezoelectric transducer
- I²C: Inter-Integrated Circuit serial communication bus
- SHM: Structural Health Monitoring
- MEMS: Micro-Electro-Mechanical Systems
- RMS: Root Mean Square vibration measure
- GPIO: General Purpose Input/Output; Wi-Fi: IEEE 802.11 wireless networking

CHAPTER 1: INTRODUCTION

1.1 Background

Bridges are critical transport assets and experience dynamic loads from moving vehicles, wind, temperature changes, and seismic activity. Structural Health Monitoring (SHM) uses measurements such as acceleration, strain, displacement, and vibration response to observe structural behaviour over time. Vibration-based methods can identify changes in response patterns that warrant further inspection. Piezoelectric transducers are useful in civil-structure monitoring because mechanical stress can generate a measurable electrical signal. MEMS accelerometers provide a compact way to measure acceleration and angular motion. Combining these sensing methods with an embedded controller and wireless communication offers a practical educational prototype for continuous data acquisition and event indication.

The proposed system uses the ESP32-C3 Super Mini as the processing and connectivity unit. The MPU6050 communicates through I²C, and the ADS1115 digitizes analog signals from two PZT sensors. The controller records sensor readings, filters noise, computes simple vibration indicators, and compares them with baseline or experimentally selected thresholds. A local indicator and optional Wi-Fi dashboard can show normal monitoring or vibration-alert states.

1.2 Problem Statement

Existing Problem: Bridge inspections are often periodic and may not capture all short-duration vibration events between visits. Visual inspection alone may not quantify dynamic response continuously.

Engineering Challenge: Bridge vibration signals vary with traffic, wind, temperature, sensor mounting, and structural configuration. A useful prototype must distinguish ordinary variation from unusually strong or changed vibration using repeatable sampling and calibrated thresholds.

Research Gap: Low-cost demonstration systems often use a single sensing method or provide raw readings without combining inertial motion measurement and piezoelectric response. Sensor fusion and clear event-state reporting can improve the educational value of a monitoring prototype.

Need for Improvement: An affordable, modular system is needed to acquire acceleration and piezoelectric signals, perform local threshold-based event detection, and make measurements available for review while clearly communicating the limitations of prototype-level monitoring.

1.3 Need of the Project

A bridge monitoring prototype can demonstrate how measured acceleration and piezoelectric response may help identify unusual vibration events that merit further investigation. An embedded, wireless system can support repeatable experiments and remote viewing of sensor trends. Earthquake-related motion detection in this project means detecting strong measured motion after shaking reaches the instrument; it is not earthquake prediction and cannot guarantee advance warning.

1.4 Objectives

The main objective is to design and test a low-cost bridge vibration monitoring prototype that combines inertial and piezoelectric sensing with embedded signal processing. Specific objectives are:

Objective 1: Interface the MPU6050 accelerometer/gyroscope and ADS1115 ADC with the ESP32-

C3 Super Mini using the I²C bus.

Objective 2: Connect two PZT sensors to ADS1115 analog inputs A0 and A1 through suitable signal conditioning and protection circuitry.

Objective 3: Acquire sensor samples, apply basic noise filtering, and calculate vibration indicators such as peak amplitude and RMS over defined windows.

Objective 4: Establish experimentally calibrated thresholds to classify normal vibration and elevated-vibration events, with local status indication and optional Wi-Fi reporting.

Objective 5: Evaluate repeatability using controlled bench/model-bridge tests and document limitations, calibration needs, and safe use.

1.5 Scope of the Project

Scope: circuit design, ESP32-C3 firmware, I²C communication, MPU6050 motion acquisition, ADS1115 conversion of two PZT signals, basic digital filtering, threshold-based event classification, data logging, and optional Wi-Fi status reporting. PZT signals may require biasing, amplification, filtering, and input protection before reaching the ADC. The prototype is not a certified structural safety system, cannot establish bridge load capacity, and must not be used as the sole basis for closing or reopening a real bridge. Future work may include more sensor nodes, calibrated reference instrumentation, long-term data storage, modal analysis, and professional validation.

Limitations: sensor placement and attachment strongly affect measurements; PZT output depends on mounting and conditioning; environmental and traffic loads can cause false alerts; and simple thresholds cannot diagnose a crack or quantify damage by themselves. Earthquake detection is limited to motion observed at the installed sensor and is not earthquake prediction.

Future Expansion: additional accelerometers and PZT channels, robust enclosures, battery backup, synchronized wireless sensor nodes, long-term trend analysis, frequency/modal analysis, and validation against calibrated engineering instruments.

1.6 Expected Outcomes

Outcome 1: A working prototype that records MPU6050 acceleration/motion and two digitized PZT sensor channels.

Outcome 2: Embedded firmware with filtering, peak/RMS calculation, baseline comparison, and threshold-based vibration event states.

Outcome 3: Local status indication and optional wireless display or logging of sensor readings and alerts.

Outcome 4: Documented circuit connections, calibration procedure, bill of materials, test results, and limitations for academic demonstration.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The literature review draws on published work on vibration-based bridge structural health monitoring, piezoelectric transducers for civil structures, seismic assessment through SHM, and the official ESP32-C3 technical documentation. Research consistently treats bridge monitoring as a combination of sensor measurement, signal

processing, environmental interpretation, and engineering assessment. Piezoelectric transducers have been studied for monitoring concrete and bridge components, while vibration-based techniques use dynamic response to assess changes in structural behaviour. The ESP32-C3 provides wireless connectivity and I²C peripheral support appropriate for a compact prototype.

2.2 Literature Review Table (Survey Table)

Table 1 summarizes representative published reviews and official technical documentation relevant to bridge SHM, piezoelectric sensing, seismic assessment, and embedded implementation.

Paper	Author & Reference	Year	Methodology / Hardware	Key Findings	Identified Research Gap
P1	Chen & Xue [1]	2018	Review of piezoelectric transducers for bridge SHM	Summarizes PZT applications in concrete and bridge component monitoring.	Mounting, environmental effects, and method selection require attention.
P2	Saidin et al. [2]	2022	Review of vibration-based bridge SHM techniques	Describes dynamic response and modal identification for bridge condition assessment.	Signals need interpretation under changing operational conditions.
P3	Karakostas et al. [3]	2024	Review of seismic assessment through SHM	Organizes damage detection, localization, quantification, and residual assessment.	A threshold alert alone cannot establish damage severity.
P4	Piezoelectric technology review [4]	2023	Active and passive piezoelectric sensing	Discusses PZT use in civil structures and hazard-related monitoring.	PZT output requires suitable mounting and signal conditioning.
P5	Zhang et al. [10]	2025	Review of bridge SHM methods and wireless monitoring	Reviews sensor methods, data processing, and early-warning applications.	Field deployment and reliable damage identification remain challenging.
P6	Espressif Systems [5]	2026	ESP32-C3 datasheet and peripheral documentation	Documents wireless connectivity, GPIO, and I²C peripheral capabilities.	Board-specific GPIO mapping must be verified.
P7	USGS [8]	2022	Earthquake early-warning system overview	Explains detection after an earthquake begins and delivery of alerts.	This prototype is not a networked, certified early-warning system.

P8	Govt. of India, MoES [9]	2021	Official statement on earthquake forecasting	Distinguishes earthquake prediction from warning after seismic activity begins.	Do not claim prediction capability.
P9	Texas Instruments [7]	Datasheet	ADS1115 16-bit I²C ADC	Provides external analog conversion for low-level sensor signals.	Input range and signal conditioning must be respected.
P10	TDK InvenSense [6]	Datasheet	MPU6050 accelerometer and gyroscope	Provides six-axis inertial measurements.	Sensor calibration and mounting affect data quality.

2.3 Research Gap

Key research observations: (1) bridge SHM benefits from vibration measurements and comparison over time; (2) piezoelectric transducers can measure stress-wave or vibration-related response but require appropriate mounting and signal conditioning; (3) seismic SHM is a staged assessment process and sensor alarms alone do not prove damage; (4) low-cost prototypes must account for noise, changing traffic/environmental loads, calibration, and false positives.

2.4 Novelty and Innovation

Combined sensing: use MPU6050 inertial measurements alongside two PZT channels to observe complementary motion/vibration signals.

External precision conversion: use ADS1115 to digitize low-level PZT outputs and provide a common I²C interface to the ESP32-C3.

Local threshold-based event detection: calculate basic vibration indicators on the controller and continue local monitoring without relying solely on a cloud service.

Low-cost modular prototype: use commonly available embedded and sensor modules for laboratory/model-bridge experiments, with calibration and safety limitations explicitly documented.

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Proposed System Architecture

The proposed architecture contains four main functional blocks: (1) sensing, comprising an MPU6050 and two PZT sensors attached to a model bridge; (2) analog signal conditioning and ADS1115 conversion for the PZT outputs; (3) ESP32-C3 Super Mini for I²C acquisition, filtering, threshold evaluation, and state management; and (4) local indication plus optional Wi-Fi-based data reporting. The MPU6050 and ADS1115 share the I²C bus; the two PZT signals connect to ADS1115 A0 and A1 through suitable conditioning circuitry.

3.2 Working Principle

The operational sequence is: (1) power up the ESP32-C3 and initialize I²C; (2) verify communication with the MPU6050 and ADS1115; (3) collect acceleration and PZT samples at a selected sampling interval; (4) filter readings and calculate peak/RMS indicators; (5) compare readings with baseline and calibrated alert thresholds; (6)

indicate normal or elevated vibration and optionally transmit readings; and (7) continue sampling and record the event for later review. Thresholds must be determined through controlled testing rather than assumed universal values.

3.3 System Architecture / Block Diagram & DFD

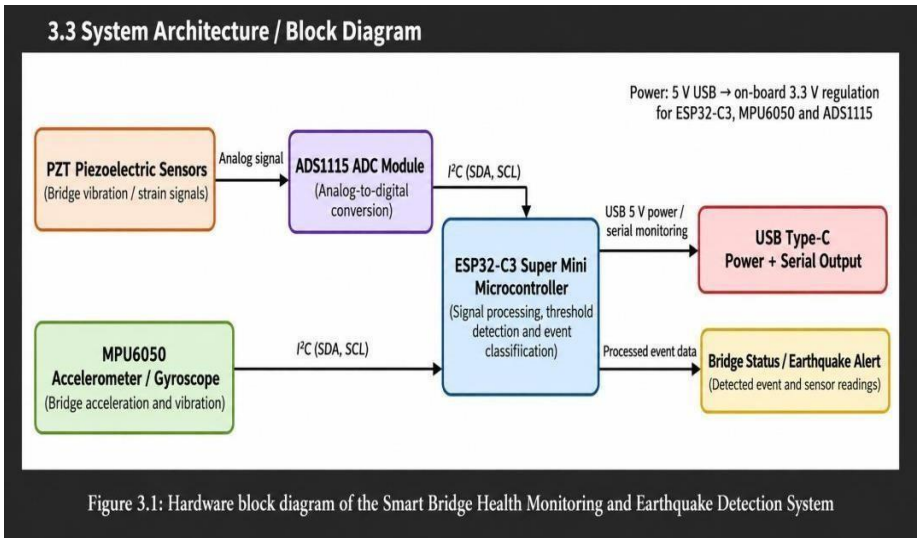


Figure 3.1: Hardware block diagram of the Smart Bridge Health Monitoring and Earthquake Detection System

3.4 Flow Chart

Figure 3 depicts the monitoring flow, including initialization, sampling, filtering, threshold decision, event indication, and return to continuous monitoring.

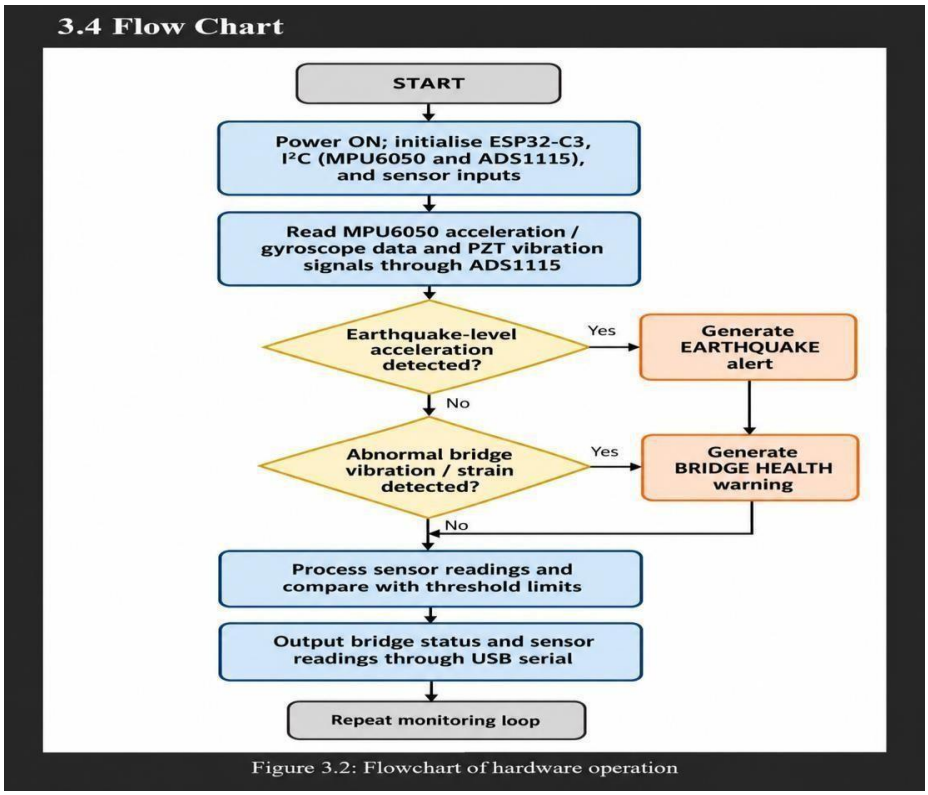


Figure 3: Flowchart of hardware operation

3.5 Methodology & UML Modeling

Figure 4 shows the circuit connection diagram, including the shared I²C bus, ADC input, power connections, and common ground.

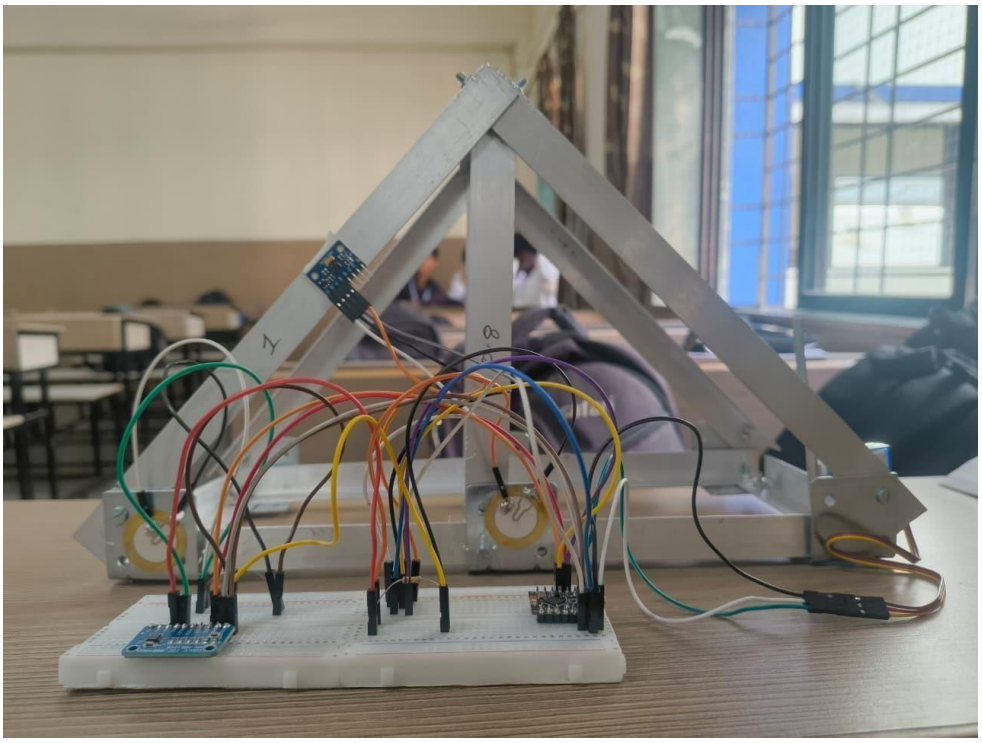


Figure 5: Prototype hardware setup on the model bridge

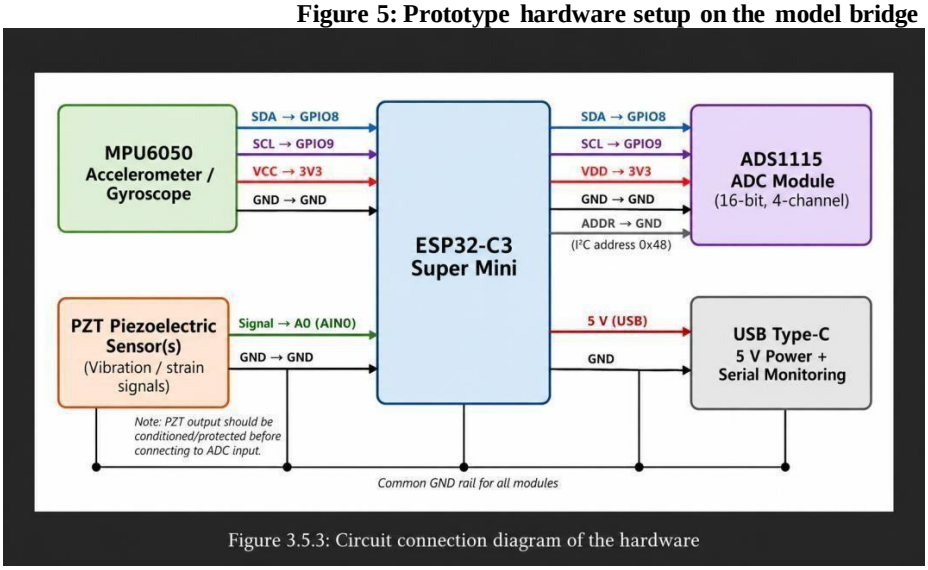


Figure 3.5.3: Circuit connection diagram of the hardware

The MPU6050 supplies digital acceleration and angular-rate measurements over I²C. The ADS1115 converts the analog PZT signals into digital samples. For a sampled channel x[n], a simple windowed RMS indicator is RMS = sqrt((1/N) × sum(x[n]²)); peak amplitude can be calculated as max(|x[n]|) over the same window. A vibration event is flagged when calibrated indicators exceed the selected thresholds for a defined duration. These indicators are screening features, not direct measurements of structural damage.

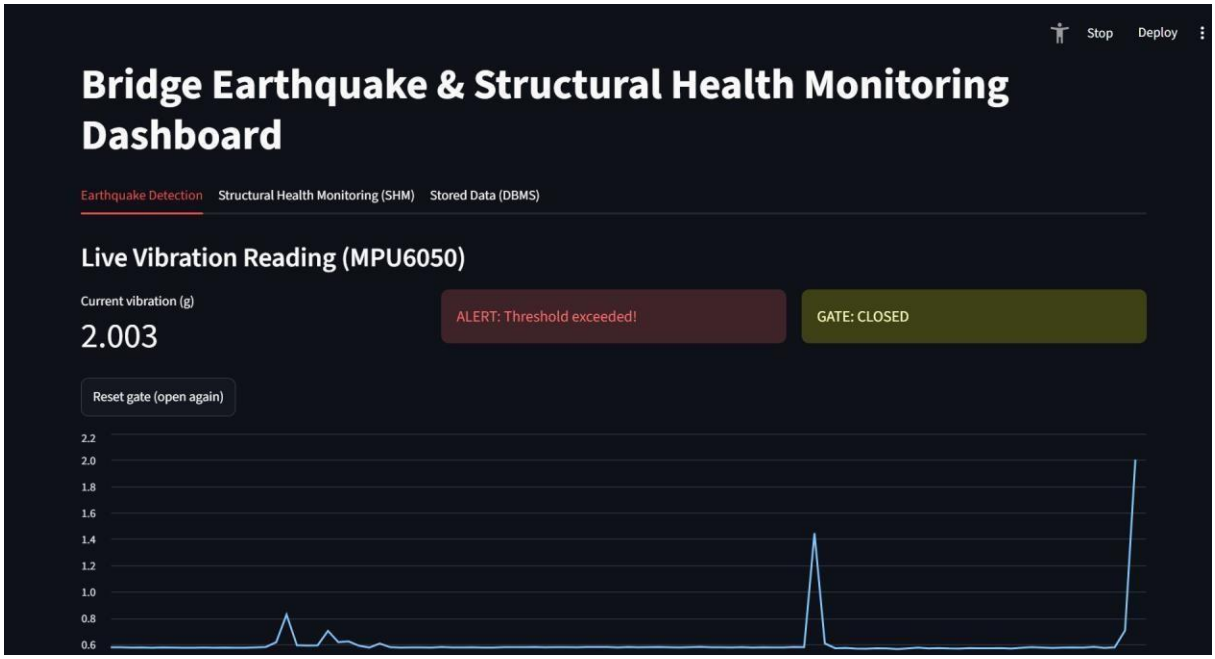
For a sampled channel x[n], a windowed RMS indicator is RMS = sqrt((1/N) × sum(x[n]²)); peak amplitude is max(|x[n]|) over the same window. A vibration event is flagged when calibrated indicators exceed selected thresholds for a defined duration. These indicators support screening only and do not directly measure structural damage.

The optional monitoring display can show the current acceleration, PZT channel readings, calculated vibration indicators, connection status, and event state. Any displayed threshold must be identified as a prototype calibration value, not a universal bridge-safety limit.

The IoT cloud web dashboard provides real-time situational awareness across three distinct operational states:

Normal Monitoring: measured values remain within the baseline range established during controlled testing.





Elevated Vibration: one or more filtered indicators exceed a calibrated threshold and an event is logged or indicated.

Recovery: readings return below the reset threshold; monitoring resumes while the event record remains available for review.

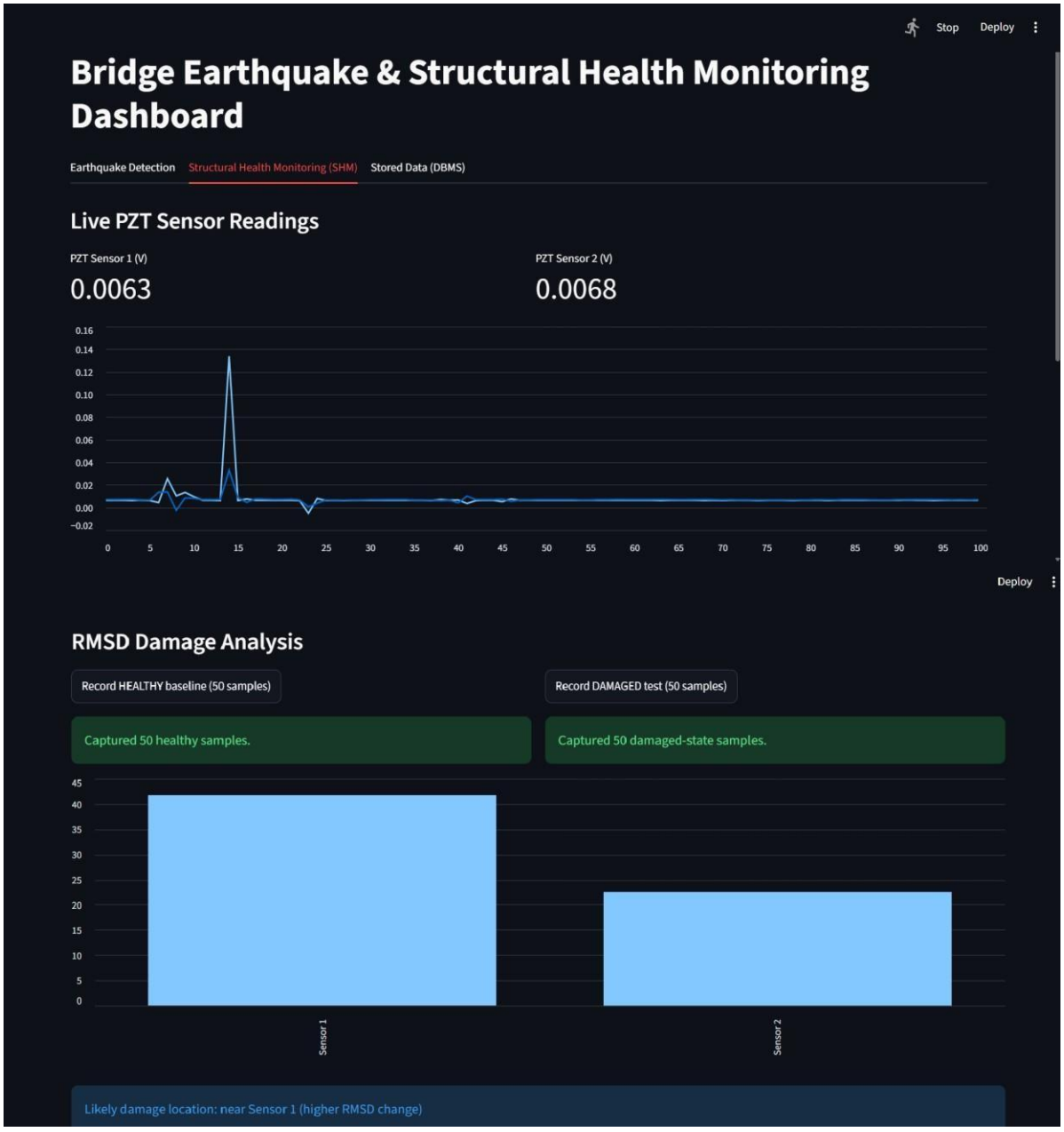
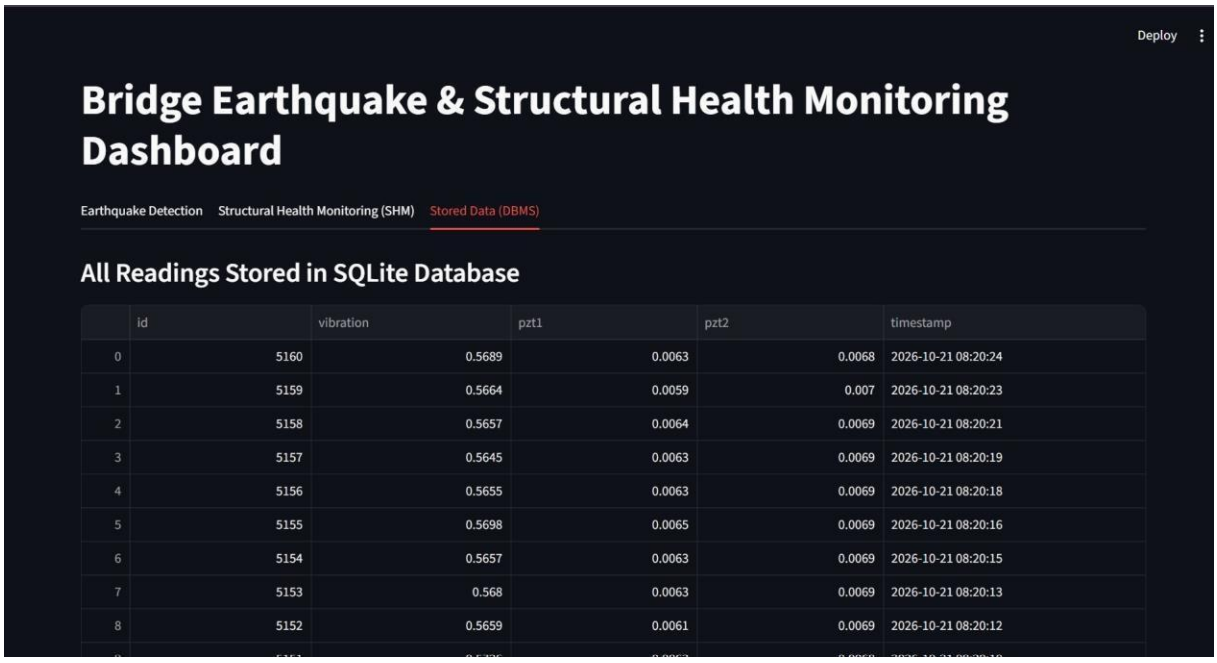


Table 2 lists the main hardware modules and their functions in the prototype.

The system should be assembled on a common ground, with the I²C bus wired according to the selected ESP32-C3 Super Mini board pinout. Use suitable PZT signal conditioning and keep all ADC inputs within the ADS1115 supply/input limits.



	id	vibration	pzt1	pzt2	timestamp
0	5160	0.5689	0.0063	0.0068	2026-10-21 08:20:24
1	5159	0.5664	0.0059	0.007	2026-10-21 08:20:23
2	5158	0.5657	0.0064	0.0069	2026-10-21 08:20:21
3	5157	0.5645	0.0063	0.0069	2026-10-21 08:20:19
4	5156	0.5655	0.0063	0.0069	2026-10-21 08:20:18
5	5155	0.5698	0.0065	0.0069	2026-10-21 08:20:16
6	5154	0.5657	0.0063	0.0069	2026-10-21 08:20:15
7	5153	0.568	0.0063	0.0069	2026-10-21 08:20:13
8	5152	0.5659	0.0061	0.0069	2026-10-21 08:20:12

The ESP32-C3 is a Wi-Fi/Bluetooth LE microcontroller with I²C peripheral support. The MPU6050 provides six-axis inertial sensing, and the ADS1115 is an external 16-bit ADC used here for two PZT channels. PZT sensors produce dynamic voltage in response to mechanical stress; their output should be conditioned and protected before digitization.

The prototype is economically feasible because it uses commonly available development boards and sensor modules. The final cost depends on local supplier pricing, mechanical mounting, signal conditioning, and enclosure choices; the budget below is an indicative estimate to be verified before purchase.

Table 2 outlines the components and technical specifications used in the prototype:

Component	Specification	Qty	Functional Role
ESP32-C3 Super Mini	Wi-Fi/BLE MCU development board; 3.3 V logic; I²C interface	1	Main controller, processing, and optional wireless reporting.
MPU6050 Module	3-axis accelerometer + 3-axis gyroscope; I²C digital output	1	Measures acceleration and angular motion of the model bridge.
ADS1115 Module	16-bit ADC; I²C; multiple analog input channels	1	Digitizes analog outputs from the PZT sensors.
PZT Sensor 1	Piezoelectric transducer; analog output requires conditioning	1	Captures local mechanical strain/vibration response.
PZT Sensor 2	Piezoelectric transducer; analog output requires conditioning	1	Provides a second vibration channel for comparison.
Signal Conditioning Circuit	Bias/protection and optional amplifier/filter as required	1 set	Keeps PZT signals within ADS1115 input limits and

			improves acquisition.
LED / Buzzer (optional)	Compatible with board logic; use driver if current exceeds GPIO limit	1 set	Local status and elevated-vibration indication.

Daily operation consists of powering the prototype, confirming sensor communication, observing baseline readings, and reviewing any elevated-vibration event. Results should be interpreted alongside the test setup and mounting conditions; alerts are prompts for inspection, not a standalone safety verdict.

4.1 Work Breakdown Structure (WBS)

Table 5 provides an indicative bill of materials. Prices are estimates and should be checked against current local suppliers before procurement.

4.2 Planning Table (Gantt Chart & Semester Timeline)

Table 2 presents the Planning Table (Semester Gantt Chart Timeline) spanning Month 1 through Month 6 of the academic year.

Project Task / Milestone	M1	M2	M3	M4	M5	M6
Literature review and component selection	■■■					
Circuit design and sensor interfacing		■■■				
Firmware and data acquisition			■■■			
Filtering, threshold logic and display				■■■		
Controlled testing and calibration					■■■	
Documentation and demonstration						■■■

4.3 Roles and Responsibilities

Table 3 outlines the distribution of technical roles and responsibilities among the four group members:

Student Name [ID / Roll No.]	Role / Designation	Specific Key Responsibilities
Jayug Nambiar [EU1258045, Roll No. 26]	System integration lead	System architecture, ESP32-C3 firmware coordination, and overall integration.
Harshit Pandey [EU1258006, Roll No. 28]	Sensor interfacing lead	MPU6050 and ADS1115 wiring, I²C checks, and sensor acquisition tests.
Nikhil Reddy [EU1258027, Roll No. 41]	PZT and signal-processing lead	PZT mounting, signal conditioning, sampling, filtering, and threshold testing.

Shayan Sayed [EU1258013, Roll No. 46]	Testing and documentation lead	Model-bridge tests, event logging, risk documentation, and report preparation.

4.4 Risk Assessment and Mitigation

Table 4 identifies potential technical and operational risks along with targeted mitigation protocols:

Identified Risk Factor	Severity & Impact	Proactive Mitigation Strategy
PZT overvoltage / transients	High: ADC damage or invalid readings	Use suitable input protection, biasing and conditioning; verify signal range before connection.
Poor sensor mounting	Moderate: inconsistent readings	Secure sensors consistently and document placement and orientation.
Traffic/environment vibration	Moderate: false alerts	Establish a baseline and require threshold exceedance over a defined sample window.
I²C wiring/address issue	Moderate: missing sensor data	Check common ground, supply, SDA/SCL mapping, addresses, and wiring length.
Power or Wi-Fi interruption	Low to moderate: remote display unavailable	Keep core sampling and local indication independent of network availability.

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

The project is technically feasible as an educational prototype. The ESP32 -C3 supports I²C communication and wireless connectivity; the MPU6050 supplies inertial measurements, and the ADS1115 provides external analog conversion for PZT signals. Reliable results depend on suitable sensor mounting, input conditioning, sampling, calibration, and repeatable tests.

5.2 Economic Feasibility

Commercial bridge SHM systems vary widely in cost depending on sensor grade, channel count, installation, communications, and analysis requirements. This project is intended as a low-cost educational prototype using readily available modules. The estimate below is indicative and excludes professional installation or certification.

5.3 Operational Feasibility

The system requires zero technical training for daily end-users. Operation is fully automated upon connection to power. Clear visual feedback (Green LED = Safe, Red LED = Alert) combined with loud acoustic sirens and explicit spoken vocal guidance ensures that all occupants—irrespective of age or literacy—can promptly comprehend the emergency and evacuate safely.

5.4 Budget Estimation & Bill of Materials (BOM)

Table 5 provides a comprehensive itemized cost breakdown and Bill of Materials for prototype construction:

Item No.	Component Description	Qty	Unit Cost (₹)	Total Cost (₹)
1	ESP32-C3 Super Mini development board	1	250	250
2	MPU6050 module	1	150	150
3	ADS1115 module	1	180	180
4	PZT sensors	2	80	160
5	Signal conditioning components / prototyping parts	1 set	250	250
6	Breadboard, jumper wires and connectors	1 set	200	200
7	LED/buzzer and resistors (optional)	1 set	80	80
8	Power cable / USB supply (if required)	1	150	150
	Indicative estimated total			1,420

Total Estimated Prototype Budget: indicative; update using actual supplier quotations

CHAPTER 6: EXPECTED RESULTS & IMPACT

6.1 Anticipated Performance Characteristics

Expected outcomes include repeatable acquisition of inertial and piezoelectric signals, detection of deliberately induced model-bridge vibration events, and logging or display of threshold crossings. Actual latency, accuracy, and false-alarm rate must be measured during testing; no performance figures are claimed before validation.

6.2 System Verification Matrix

Table 6 describes expected system responses for selected test conditions. These are test cases to validate, not guaranteed performance results.

Operating State	MPU6050 State	PZT / ADS1115 State	Local Response	Optional Data Status
State 1: Normal	Acceleration within calibrated baseline	A0/A1 below threshold	Normal indicator; continue sampling	NORMAL
State 2: Elevated Vibration	Peak/RMS indicator exceeds threshold	One or both PZT channels elevated	Alert indication; log event	VIBRATION ALERT
State 3: Multi-sensor Event	Elevated inertial response	PZT channel(s) elevated in same window	Higher-priority alert for review	POSSIBLE STRONG MOTION
State 4: Sensor/Link	No valid or stale	ADC/I²C read	Show fault; do not label	SENSOR FAULT

Fault	readings	error	as normal	
State 5: Offline Mode	Any State	Any State	Local Alarms Fully Operational; Failsafe Active	Status: OFFLINE (Local Buffer)

6.3 Safe Simulation and Testing Protocol

Testing should use a small model bridge or bench fixture and controlled tapping/vibration sources. Do not perform unsafe loading or induce damage to a real bridge. Compare results with repeatable reference conditions, secure the sensors, inspect wiring, and record thresholds and sampling settings. Earthquake-related alerts indicate detected motion only and must not be presented as prediction or certified early warning.

6.4 Industrial Relevance & Societal Impact

Bridge owners and maintenance teams use structural health monitoring to support condition assessment, maintenance planning, and post-event inspection. This educational prototype demonstrates the sensor-acquisition and event-indication concepts but is not a substitute for professional structural assessment or code-compliant monitoring systems.

6.5 UN Sustainable Development Goals (SDG) Mapping

The project directly maps to three United Nations Sustainable Development Goals (SDGs):

- ☑ **SDG 9: Industry, Innovation and Infrastructure — demonstrates a low-cost educational approach to sensing and infrastructure monitoring.**
- ☑ **SDG 11: Sustainable Cities and Communities — supports awareness of infrastructure resilience and the value of monitoring critical assets.**
- ☑ **SDG 13: Climate Action — contributes to learning about resilient infrastructure and disaster-risk awareness; it does not itself prevent or predict earthquakes.**

CHAPTER 7: CONCLUSION

7.1 Summary of the Project

The Smart Bridge Health Monitoring and Earthquake Detection System combines an ESP32 -C3 Super Mini, MPU6050, ADS1115, and two PZT sensors to monitor vibration in a model bridge. Inertial readings and digitized piezoelectric signals are sampled, filtered, and compared with experimentally calibrated thresholds. When elevated vibration is detected, the system can indicate and optionally report the event for further review. The prototype demonstrates a practical embedded SHM workflow while recognizing that real bridge safety decisions require validated instrumentation and qualified engineering assessment.

7.2 Key Contributions

The main contributions are: (1) integration of inertial and piezoelectric sensing; (2) I²C acquisition through the ESP32-C3 and ADS1115; (3) simple filtering and threshold-based event indication; and (4) a documented hardware connection plan and controlled test methodology.

7.3 Future Scope

Future enhancements may include more PZT/accelerometer channels, improved analog front ends, calibrated seismic sensors, long-term cloud logging, frequency-domain and modal analysis, sensor synchronization, robust outdoor packaging, and validation on engineering test structures. Earthquake prediction is outside the project scope.

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APPENDICES (Optional)

Appendix A: Hardware Interfacing and Pinout Specification

Table A1 details the proposed pin interconnections for the ESP32-C3 Super Mini, MPU6050, ADS1115, and two PZT sensors. Confirm the exact SDA/SCL labels and GPIO mapping printed on your specific board before wiring.

Module / Device	Device Pin	ESP32-C3 Super Mini / ADC Pin	Signal Type / Function
MPU6050	VCC / GND	3V3 / GND	Power; use common

			ground.
MPU6050	SDA / SCL	GPIO8 SDA / GPIO9 SCL (proposed)	Shared I²C bus; confirm board pin labels.
ADS1115	VDD / GND	3V3 / GND	Power and common ground.
ADS1115	SDA / SCL	GPIO8 SDA / GPIO9 SCL (shared)	Same I²C bus as MPU6050; verify address does not conflict.
PZT Sensor 1	Signal / return	ADS1115 A0 through conditioning	Analog vibration signal; never exceed ADC input limits.
PZT Sensor 2	Signal / return	ADS1115 A1 through conditioning	Second analog vibration signal; never exceed ADC input limits.
Optional LED/Buzzer	GPIO input/driver	Available GPIO through resistor/driver	Local status; do not exceed GPIO current limits.

Appendix B: Proposed Data Record

Example data fields for local logging or optional Wi-Fi reporting: timestamp; MPU6050 acceleration (ax, ay, az); angular rate (gx, gy, gz); ADS1115 PZT channel A0; ADS1115 PZT channel A1; filtered peak/RMS values; event state; threshold configuration; and sensor/communication status. Sample values should come from actual tests and must not be fabricated.